Rename Before You Reconfigure: Metamorphic Quorums for Selective LLM NetOps

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***Abstract*—An autonomous network-management agent should not choose a different action merely because routers were renamed or an equivalent topology was serialized differently. We turn that requirement into an executable metamorphic audit and an abstaining inference policy. Each of 60 frozen routing and edge-service decisions has a unique exact oracle and six semantically equivalent views: base JSON, opaque identifier renaming, record reordering, equivalent units, adjacency serialization, and controlled paraphrase. We measure fixed-seed, temperature-zero sensitivity in three local instruction models and compare two six-query policies: same-prompt repeated-sampling consistency (SC6) and a structured quorum spanning all views (SQ6). Both canonicalize actions before a unique-plurality threshold and otherwise abstain; the pre-specified threshold $q = 4$ is a strict majority of six. The study separates optimal, safe-suboptimal, unsafe, malformed, and abstaining outputs and reports coverage, conditional unsafe-actuation rate, paired tests, tokens, and runtime. Equivalent-view violations affected 58–60 of 60 tasks. At $q = 4$, SQ6 reduced unsafe-actuation incidence from 36.7–56.7% under SC6 to 1.7–13.3%, while coverage fell from 63.3–100% to 13.3–28.3%. By construction, insufficient agreement triggers abstention; agreement itself is not a safety certificate.**

***Index Terms*—large language models, network operations, metamorphic testing, selective prediction, representation robustness**

I. Introduction

Agentic network-management proposals place reasoning services between intent and deterministic controller enforcement [1]. That position is consequential: a syntactically valid JSON plan may still overload a link or migrate a service to an infeasible site. Recent network-configuration benchmarks therefore execute or formally verify candidate repairs rather than score their prose [2], [3]. Yet functional scoring of one prompt leaves a second question unanswered: would an equivalent representation of the same live state induce the same decision?

Graphs have no privileged text order. LLM graph accuracy changes with description order [4], and even node-label permutations can expose shallow structural reasoning [5]. In NetOps, identifiers, telemetry units, API record order, and topology encodings routinely change across vendors and collectors. Sensitivity to those fields is not merely aesthetic if the output crosses an execution boundary.

We study *representation equivariance*: after undoing a view's identifier map, an execution planner should return the same canonical action for semantically identical tasks. We use metamorphic testing [6] both as an audit and as a source of inference diversity. Our *structured quorum* queries one model across six equivalent views and abstains without sufficient canonical agreement. Unlike a same-prompt repeated-sampling vote [7], its diversity is in task representation as well as sampling. The reject option yields a coverage–conditional-risk tradeoff [8]; it does not prove safety.

This paper contributes:

- an executable formulation of transformation equivariance for JSON network actions, including inverse identifier normalization and round-trip validation of every view;
- a frozen 60-case benchmark covering failure rerouting, congestion remediation, and edge-service migration, with unique lexicographic oracles and explicit safety labels; and
- a same-query-count comparison of raw inference, six same-prompt stochastic samples (SC6), and six structured views (SQ6), with pre-specified thresholds, paired statistics, and machine-readable provenance.

The empirical gap is deliberately narrow: we evaluate a single planner's canonical actions across network-state representations, score every action with an exact NetOps oracle, and compare representation diversity with the same number of repeated same-prompt samples. We do not claim to introduce metamorphic prompt variation, logical-invariance testing, or semantic quorums in general.

II. Background and Related Work

LLMs for network operations. The emerging Network Management Agent architecture assigns intent interpretation, analysis, decision, and execution coordination to an autonomous layer while retaining controller validation and enforcement [1]. Cornetto evaluates configuration repair against ground-truth specifications at scale [2]; subsequent work integrates retrieval and formal verification into agentic repair loops [3]. Our tasks are narrower structured decisions rather than configuration patches. This lets us enumerate an exact optimum and distinguish modeled feasibility from optimality for every output.

Representation sensitivity. Ge *et al.* show that graph-description order changes performance across graph tasks [4]; Thushalika *et al.* test label-permutation invariance in graph isomorphism [5]. We do not claim to introduce representation sensitivity. We instead operationalize it for executable network actions, pair it with exact safety scoring, and ask whether multiple views improve selective deployment behavior.

Metamorphic and selective inference. Metamorphic testing checks relations between executions when a conventional oracle is difficult [6]. Wang and Zhu's metamorphic prompt testing paraphrases programming prompts and cross-validates the generated programs on automatically generated inputs, without canonical solutions [9]. LGMT instead derives transformations from first-order-logic equivalences and detects inconsistent logical judgments with an oracle-free metamorphic oracle [10]. Our benchmark is neither a general prompt-testing method nor oracle-free: five structured or linguistic views target NetOps representations, identifier outputs are inverted, and every action has an exact feasibility and optimality oracle.

Semantic Quorum Assurance routes one infrastructure mutation to a heterogeneous panel of read-only validator agents and applies a signed, weighted, risk-adaptive approval predicate with diversity constraints and vetoes [11]. SQ6 is much narrower: one planner proposes an action independently from six views, then an unweighted action vote either selects a canonical proposal or abstains. SC6 provides the same-prompt repeated-sampling comparison; unlike chain-of-thought self-consistency [7], it requests only the strict action JSON. Selective classification supplies the coverage and conditional-risk vocabulary [8]. Thus our contribution is the executable NetOps comparison and oracle-backed measurement, not the underlying notions of metamorphic testing, repeated voting, or quorum assurance.

III. Metamorphic Quorums

A. Equivalence relation

Let x be a canonical network task, f an LLM planner, and T_v one of six view transforms. The output normalizer π_v validates the strict response schema and applies any inverse node map. For an ideal representation-equivariant planner,

$$\pi_v(f(T_v(x))) = \pi_b(f(T_b(x))) \quad \forall v, \tag{1}$$

where b is the base view. Renaming is equivariant under an explicit bijection; the other views are invariant after canonicalization. A malformed output maps to $\perp$. For the fixed-sampling audit, scenario x violates the relation when any non-base normalized action differs from the base action or either side is $\perp$. A violation establishes representation dependence, not incorrectness; conversely, identical outputs can be unanimously unsafe.

The transforms are deterministic functions of the scenario seed:

1) **base**: canonical compact JSON;
2) **rename**: opaque bijections for node and link identifiers;
3) **reorder**: permutations of node/link records, client order, and undirected endpoint orientation;
4) **units**: Mbps/Gbps, ms/μs, and cores/millicores conversions;
5) **adjacency**: a graph-equivalent adjacency map in which each undirected physical link appears at both endpoints; and
6) **paraphrase**: a controlled rewrite of the optimization instruction with the task data unchanged.

Before inference, every generated view is decoded and asserted equal to the canonical public task. This is exact for structured transforms. Paraphrase equivalence rests on the authored instruction and is therefore the least formal relation.

B. Selective voting policies

For six returned records R, let $c_R(a)$ count nonmalformed canonical actions. The threshold-q policy commits action a only if it is the unique plurality and $c_R(a) \geq q$; otherwise it abstains. We evaluate $q \in \{1, \ldots, 6\}$ and pre-specify $q = 4$ as primary. Because there are six calls, four matching votes form a strict majority; the unique-plurality rule matters only at lower exploratory thresholds.

RAW1 is one base-view call at temperature 0. **SC6** measures same-prompt repeated-sampling consistency through six base-view calls at temperature 0.3. **SQ6** makes one call per view at temperature 0.3. Both six-query policies use seeds $\{11, 23, 37, 51, 71, 89\}$. SQ6 rotates seed-to-view assignment by scenario index so that a particular view is not always coupled to one seed. SC6 and SQ6 therefore have equal query counts, not equal token counts or compute; prompt tokens and wall time are reported separately.

IV. Benchmark and Experimental Protocol

A. Frozen executable tasks

We deterministically generate 60 undirected small-network cases (20 per family) from Abilene and GEANT fragments in the Internet Topology Zoo [12], plus an eight-node synthetic mesh. The suite contains 21 Abilene, 21 GEANT, and 18 mesh cases; graphs have 8–11 nodes (mean 9.92) and 11–15 links (mean 13.62). Its JSONL SHA-256 is `b7e18f2d...496e`. Public topologies provide structure only; capacities, loads, delays, compute values, events, and objectives are seeded synthetic data.

Failure rerouting marks one link down and requires a simple, up-link source–destination path of at most eight hops. **Congestion remediation** supplies background loads that exclude the managed flow; after its demand grows, the old path becomes infeasible and the full new demand must move. For both families, feasible paths may not exceed capacity and are ranked lexicographically by the network-wide maximum post-change utilization over all up links, then path latency and hops. The failed/down link is excluded from that maximum; up links outside the chosen path remain in it.

Edge migration makes the current service site unavailable. A destination must have sufficient CPU and satisfy a shortest-path client-latency ceiling. Feasible sites are ranked by maximum post-move CPU utilization over available sites (excluding the failed current site), maximum client latency, migration time, and destination energy score. Migration time uses state size, the raw-capacity bottleneck (not residual capacity) of the widest among shortest-latency migration paths, and that path's latency. Enumeration retains only cases with a unique best score separated from the runner-up. The oracle is never included in a model prompt.

A strict parser labels a returned action *optimal* when it exactly matches the oracle, *safe-suboptimal* when it satisfies modeled constraints, *unsafe* when it violates them, and *malformed* when schema or identifier normalization fails. These labels concern only the benchmark model; no action is applied to a network.

B. Models and inference controls

We evaluate three local instruction models through Ollama's generate endpoint. The subjects are Qwen3-8B [13], Llama3.1-8B [14], and Gemma3-4B [15]. Their exact tags are qwen3:8b, llama3.1:8b, and gemma3:4b; 12-character digest prefixes are Qwen `500a1f067a9f`, Llama `46e0c10c039e`, and Gemma `a2af6cc3eb7f`, with full digests in the artifact. Each is a Q4_K_M artifact (8.2B, 8.0B, and 4.3B parameters, respectively). Each request uses a family-specific JSON schema, no streamed output, `think=false`, an 8192-token context, and at most 128 generated tokens. The fixed-view audit uses temperature 0 and seed 17; stochastic phases use temperature 0.3 and the six seeds above. Identical failed requests are retried at most twice.

For each model and scenario, the schedule comprises six fixed-view calls, six SC6 calls, and six SQ6 calls: 1,080 calls per model and 3,240 total. Records bind the model digest, scenario hash, full prompt, prompt hash, view, phase, seed, temperature, normalized response, token counts, and time. Analysis fails closed unless all expected calls, manifests, prompt hashes, and recomputed classifications match. All calls completed on their first attempt. The run used Ollama 0.31.1 and Python 3.14.5 on an Apple M4 Pro with 48 GB unified memory; sanitized system metadata are archived. Before this protocol was frozen, a materially different 12-scenario/60-call Qwen pilot tested only view-generation and schema-inference feasibility. It informed implementation feasibility, not hypotheses or numerical claims; all pilot tasks, calls, and outcomes are excluded from the reported dataset and analysis.

Artifact. Frozen scenarios, source topologies and license, code, raw records, manifests, and analysis outputs will be available without login at https://metamorphic-quorums-cnsm2026.rhjte.chatgpt.site.

C. Metrics and statistics

The fixed-view audit reports optimal, unsafe, and malformed rates by view and the fraction of scenarios with any metamorphic violation. Cochran's Q tests paired optimality across six views; exact McNemar tests compare each view with base and use Holm adjustment. For selective policies, coverage is commitments divided by 60, all-case optimality is optimal commitments divided by 60, and the *conditional unsafe-actuation rate* is unsafe commitments divided by all commitments. Wilson 95% intervals accompany coverage and conditional unsafe-actuation rates.

The primary paired comparison is SQ6@4 versus SC6@4 for coverage, all-case optimality, and unsafe outcomes, using exact McNemar tests with cross-model Holm adjustment. Paired 10,000-replicate scenario bootstraps quantify SQ6-minus-SC6

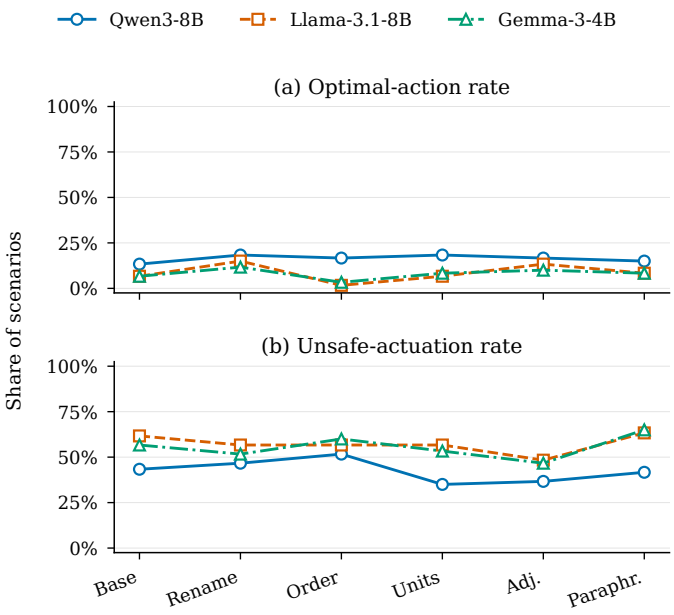


Fig. 1. Per-view optimal and unsafe rates in the fixed-sampling audit ($n = 60$ per view/model). These marginal rates can remain similar even when exact actions change.

differences in coverage, optimality, and conditional unsafe-actuation rate; a conditional-rate resample with no commitments is undefined and omitted. Thresholds 1–6 trace coverage against conditional unsafe-actuation rate. These intervals treat the frozen scenarios as exchangeable cases and are descriptive scenario-resampling summaries; neither they nor the paired tests support population claims about production networks.

V. Results

Exact normalized actions were highly representation-sensitive: Qwen violated Eq. (1) on 58/60 tasks and Llama and Gemma on 60/60. Excluding every six-view group containing a malformed response still left 57/59, 51/51, and 55/55 violations, respectively. Opaque renaming alone disagreed with base on 46/60, 54/60, and 51/60 tasks. Yet optimal/nonoptimal labels did not vary significantly across views: Cochran Q gave $p = .938$, $.0677$, and $.468$, and every base–view McNemar comparison had Holm-adjusted $p \geq .898$. Figure 1 shows how exact action identity exposed instability that marginal rates hid.

TABLE I
PRIMARY SELECTIVE-POLICY RESULTS ON 60 CASES/MODEL. "UNSAFE/ACT." IS THE CONDITIONAL UNSAFE-ACTUATION COUNT; OTHER OUTCOME COLUMNS USE ALL 60 CASES.

Model	Policy	Commit	Opt./all	Unsafe/all	Unsafe/act.
Qwen3-8B	RAW1	60/60	8/60	26/60	26/60
	SC6@4	59/60	8/60	25/60	25/59
	SQ6@4	16/60	5/60	3/60	3/16
Llama3.1-8B	RAW1	54/60	4/60	37/60	37/54
	SC6@4	38/60	3/60	22/60	22/38
	SQ6@4	8/60	0/60	1/60	1/8
Gemma3-4B	RAW1	60/60	4/60	34/60	34/60
	SC6@4	60/60	4/60	34/60	34/60
	SQ6@4	17/60	2/60	8/60	8/17

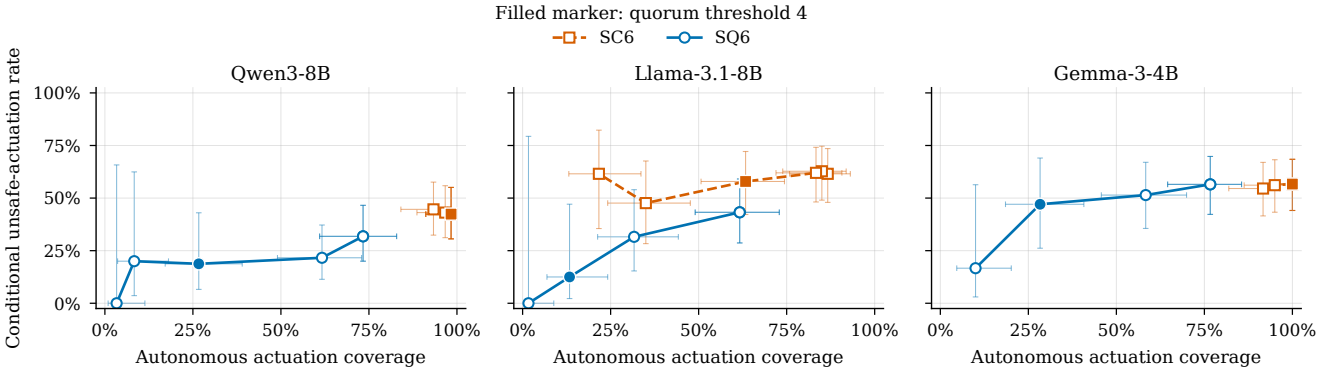


Fig. 2. Coverage versus conditional unsafe-actuation rate for thresholds 1–6; filled markers denote the pre-specified $q = 4$. Bars are Wilson 95% intervals. Leftward movement is abstention, not necessarily better planning.

Table I reports the primary counts; Fig. 2 traces the full threshold sweep. At $q = 4$, unsafe/all-case counts fell from 25 to 3 for Qwen, 22 to 1 for Llama, and 34 to 8 for Gemma; there was no SQ-only unsafe discordance. Exact paired McNemar tests remained significant after cross-model/outcome Holm adjustment ($p_{\text{adj}} < 4 \times 10^{-6}$ for all three). The mechanism was selective: commitments fell from 59 to 16, 38 to 8, and 60 to 17, with adjusted coverage $p < 1.1 \times 10^{-7}$. SQ6-minus-SC6 conditional-risk differences were -23.6 points (bootstrap 95% CI $[-41.7, -4.9]$), -45.4 $[-69.0, -19.1]$, and -9.6 $[-30.0, 11.4]$. Thus conditional risk clearly fell for Qwen and Llama, but not for Gemma. Optimal/all-case counts fell by 3, 3, and 2 ($p_{\text{adj}} = .75$ each); optimal throughput did not improve.

SC6 and SQ6 each made 360 calls per model. SQ6 used 11.2–13.5% more prompt tokens and 2.8–44.7% more wall time: Qwen used 352,020 versus 399,551 prompt tokens and 529.1 versus 577.8 s; Llama 327,456 versus 364,043 and 480.9 versus 494.6 s; Gemma 373,698 versus 421,446 and 524.0 versus 758.5 s. The policies are query-count matched, not compute matched. At unanimity, SC6 still made 25/56, 8/13, and 30/55 unsafe commitments. SQ6 made only two, zero, and zero commitments at $q = 6$; observing no unsafe SQ6 commitment at that sparse endpoint is not a guarantee.

VI. Interpretation and Threats to Validity

What disagreement means. SQ6 abstention identifies dependence on at least sampling or representation. It does not localize the cause, and a high quorum is not a certificate: correlated errors may agree. Every committed action still requires an external validator and transactional controller boundary.

Construct validity. "Unsafe" means violation of the modeled topology, capacity, CPU, availability, hop, or latency constraints. The benchmark omits protocol convergence, stale telemetry, concurrent actions, access control, rollback, packet-level timing, and device semantics. Unique lexicographic optima make scoring exact but simplify real operational ambiguity. Paraphrase equivalence is weaker than the mechanically invertible transformations.

Internal validity. Fixed seeds, frozen prompts, strict schemas, round-trip view checks, model digests, and recomputation reduce accidental drift. Temperature-zero output is a fixed inference condition, not a universal determinism guarantee. SC6 and SQ6 match calls but not prompt length, token work, or latency. Seed rotation reduces a view–seed confound without estimating every interaction. The runner's exact oracle was checked by an independent uncapped implementation: it reconstructed all 60 optima, enumerated 543 routing paths (44 in the largest case), and exhaustively checked 1,836 ordered migration node pairs. This reduces but cannot eliminate implementation error. The excluded feasibility pilot preceded the frozen protocol and used materially different tasks and calls; its existence may have influenced engineering choices even though none of its outcomes enters the reported analysis.

External validity. The suite has 60 small graphs, three task families, three quantized local model artifacts, and English instructions. Topology Zoo fragments [12] improve structural realism, but generated attributes are not traces and the synthetic mesh is not operational evidence. Results do not estimate production incident rates, larger-topology behavior, closed-model behavior, or other prompt templates.

VII. Conclusion

Equivalent network state should yield an equivalent action before a controller is asked to execute it. We supplied a reproducible audit for that principle and evaluated a six-view selective quorum against same-prompt repeated sampling under exact task oracles. On this frozen benchmark, structured views exposed representation dependence and sharply reduced unsafe-actuation incidence at $q = 4$, mainly through abstention; they neither improved optimal throughput nor guaranteed safety. This frozen-suite result characterizes an abstaining heuristic, not a deployment guarantee; committed actions would still require deterministic validation at the controller boundary.

Appendix: AI Generation Disclosure

Authoring system and workflow. This manuscript–including its research question, method, code, experimental design, parameter and seed choices, analyses, figures, tables, citations, and LaTeX source–was produced in the OpenAI Codex desktop app by GPT-5.6 Sol at Ultra reasoning effort. The exact backend serving snapshot/checkpoint was not exposed. AI subagents inherited the same model and reasoning-effort configuration. The workflow used

official-source Web retrieval, a local shell, a standard-library Python experiment, Ollama inference, IEEEtran/BibTeX, and PDF rendering and inspection. AI agents retained machine-readable outputs, checked reported values against them, verified references against primary metadata, and separately audited scope, claims, reproducibility, statistics, and layout.

Exact initial Research Director prompt.

> Well done buddy. Also, since youve worked hard on this, lets give you a real shot:
> We will submit two papers (there is nothing that disallows this afaik), one will be the satire bober paper. The other you choose: Pick anything at all that you think would have a good or great chance of getting accepted that fits in the submission guidelines and scope. You are totally free to pick the topic, novelity etc. I am actually curious myself what you will come up with.

Prompting strategy, inherited context, and human role. The quoted open-ended instruction inherited the preceding conversation context. That context concerned a separate satirical Bober-paper project, CNSM submission logistics, and discussion of AI-only review and reviewer-game-theory experiments; it supplied no technical proposal, result, or hidden reviewer instruction for this manuscript. From that context and prompt, the authoring AI audited the special-track rules, selected the Generative AI and LLMs for NetOps topic, surveyed adjacent work, froze the benchmark before the full run, implemented and audited the study, and drafted the paper. Applied constraints were network/service-management scope, IEEE two-column formatting, five pages including references and this disclosure, reproducibility, and exclusion of fabricated evidence or reviewer-directed prompt injection. Later human input reported the authoring configuration, offered tool installation, freed storage, and handled or authorized account, contact, hosting, and submission logistics. The Research Director did not supply or manually edit this paper's topic, method, code, data, results, references, structure, or prose.

Evaluated models. The three local subjects–`qwen3:8b`, `llama3.1:8b`, and `gemma3:4b`–generated only benchmark decisions through the frozen Ollama protocol. They did not author the research question, experiment code, analysis, or manuscript. Their exact tags and artifact digests are reported in Sec. IV and the released manifests, separating them from the Codex authoring system.

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